UFCEP4-0-3 · BUSINESS PROCESS MODELLING AND ENTERPRISE ARCHITECTURE

Hospital Patient Referral, Treatment & Administration System

The operational model & measurable acceptance criteria

Two required deliverables today

1. **The operational BPMN diagram — O1_Operational_Merged**
2. **The measurable acceptance criteria for that model**

Group: max · finn · henry · peter

28 September 2026

finn

Two deliverables, four speakers

PART A · The operational model

finn — slides 3-4

- the whole O1 diagram
- how to read it: boundary, responsibility, communication

henry — slides 5-6

- the 12 decision gateways = the business rules
- separation of duties + validation evidence

PART B · Measurable acceptance criteria

peter — slides 7-8

- why the original 20 criteria were not measurable
- the revised 28, and four worked examples

max — slides 9-10

- execution evidence and the honest limits
- conclusion and discussion points

Ten slides, about ten minutes. Anything beyond the two required artefacts is kept to one evidence slide and two backup slides.

Backup slides: A · the fifteen message flows (11) B · how the executable part is put together (12)

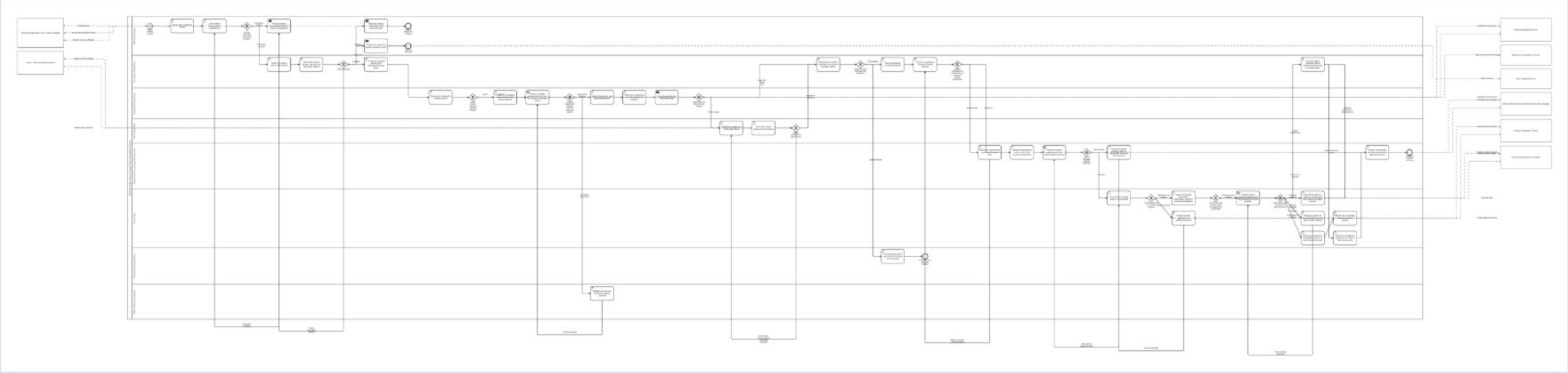
peter and max each take two slides, and the last two slides are deliberately short.

finn

A1 The operational model — O1_Operational_Merged

From referral intake to a confirmed treatment schedule — one collaboration.

The complete operational model — 8 hospital swimlanes, 8 external black-box pools, 15 labelled message flows



8 swimlanes (hospital teams)

9 pools: 1 hospital + 8 external

15 labelled message flows

36 activities · 12 decision gateways

finn

A2 Three ideas: boundary, responsibility, communication

1 Boundary

- one white-box pool: the hospital
- eight black-box pools: external parties
- the case study does not describe their internals, so we do not invent them

2 Responsibility

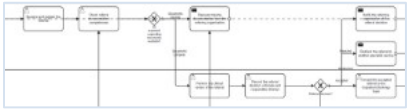
- eight lanes = the eight accountable teams
- clinical judgement only in clinical lanes
- money only in the Finance lane

3 Communication

- 15 named message flows, each with a source and a target
- four request / response pairs
- the register is on backup slide A

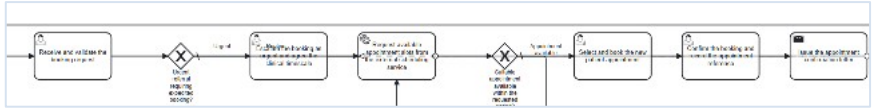
Intake and the clinical decision

Missing documentation returns to the completeness check; accepted, rejected and redirected each have their own end event and message flow.



Booking and the external scheduling service

No suitable slot → escalate to the Pathway Coordinator; an urgent referral is expedited before the slot query.



Every gateway has a condition and a default branch, so the process can never stall on a missing outgoing flow.

henry

A3 The twelve decision gateways are the business rules

Documents complete?	missing → request from the referrer, then loop back to the check
Referral decision?	accepted / rejected / redirected / more information — four controlled outcomes
Urgent referral?	expedite the booking before the normal slot query
Slot available?	book, or escalate to the Patient Pathway Coordinator
Within 14 days?	telephone the patient, or no call at all
Contact confirmed?	booking stands, or the failed attempt is recorded
Consent given?	record consent, or close the pathway as declined
Request authorised?	book the treatment, or return it to the clinical team
Capacity confirmed?	confirm the schedule, or keep pending and record the retry
Funding approved?	continue, or resolve the funding before confirming
Advance payment?	secure payment request, or confirm without a charge
Payment outcome?	confirmed / declined (one retry) / unconfirmed → investigate, never re-request

Each gateway has a controlled value set and a default branch: the engine can always choose a path.

henry

A4 Separation of duties, and proof the model runs

Clinical

- only a consultant accepts, rejects, redirects or judges suitability
- consent is recorded before any treatment booking
- no worker takes a clinical decision

Administrative

- register, check documents, request missing information
- book, confirm, issue letters, record contact attempts
- may not change clinical meaning

Financial

- funding route, secure payment, refunds, investigations
- card data is never stored
- a clinical decision is not a payment authority

0 moddle warnings

0 Camunda 8 lint reports

6 / 6 definitions deployed

41 worker unit tests

- Deploying to a real Camunda 8 cluster found four defects that no diagram review had found — three XSD sequence violations and one runtime form-binding defect. All four were fixed and re-verified.
- Honest caveat: candidate groups express intent, but the engine does not enforce group membership in this build — one of the unmet criteria on slide 9.

peter

B1 Rewriting the criteria: 20 → 28 measurable ones

Five problems with the original twenty

1 they described activities, not outcomes
2 only two carried a threshold, so eighteen could not fail
3 nine exception requirements had no criterion at all
4 non-functional needs were one unmeasurable line
5 wording like "handled" or "properly" is not verifiable

What we changed

- each criterion is a terminal state or a persisted record
- thresholds added where the case study gives one (14 days, 7 days, amounts)
- the missing exception paths written in
- every criterion names a variable, a model activity and a test case
- twelve keep the original wording alongside the revision

11 normal path

9 exception path

5 business rule

3 non-functional

Every criterion is written as Given … when … then …, or as a measurable statement, and traces to a requirement (REQ-nn), a business rule (BR-nn) and a model activity.

peter

B2 Four worked examples

AC-R-06 14-day rule

Given an appointment within 14 days, then a telephone task is created for the Call Handling Team; more than 14 days, then no call.
REQ-09 · BR-02 · TC-06, TC-07

AC-R-11 declined payment

Given a declined payment, then the patient and the team are notified and one further attempt is allowed — with no second appointment and no second charge. REQ-22 · BR-05 · TC-15

AC-R-12 unconfirmed payment

Given a payment taken but not confirmed, then the transaction is marked for investigation and no second payment request is raised.
REQ-23 · BR-06 · TC-16

AC-R-19 card data

Given any payment channel, then no complete card number or security code is stored. REQ-21 · CN-01 · design inspection — no test case, and we say so.

Each criterion has the three parts that make it measurable: a precondition, an action, and an observable result.

max

B3 What we verified, and what we do not claim

18 / 18 cases passed

0 incidents

6 definitions deployed

build 1.0

Verified

- 18 in-scope cases executed against build 1.0 in one run
- a case passes only if the expected elements were reached, nothing that should have been skipped was reached, the final state matched and no incident was raised
- external services were exercised at the process boundary

Not met — reported as gaps

- RBAC enforcement and immutable audit records
- management reporting output (the monthly timer has not run)
- live integration with any external provider
- availability / downtime — no procedure to test

The execution run is evidence, not a requirement: the brief asks for the criteria, and the criteria are on the previous two slides.

max

What we claim, and what we do not

We claim

- the operational model is complete, clean and executable
- every external exchange is an explicit message flow
- the acceptance criteria are measurable and traceable
- the referral-to-schedule path runs with no incidents

We do not claim

- enforced RBAC or an immutable audit store
- live external integrations
- management reporting output
- outage behaviour

Discussion points

1. Should management reporting be in the first release?
2. How far should external integration go in release 2?
3. Is lane-based separation enough without enforced RBAC?

Thank you — questions welcome. Backup: A · the fifteen message flows (11) B · the executable part (12).

Backup

A The fifteen message flows, and who they connect

MF_01	Referring organisation → hospital: Patient referral
MF_02	hospital → referring organisation: Missing documentation request
MF_03	hospital → referring organisation: Referral outcome notification
MF_04	hospital → other specialist service: Referral redirect
MF_05	hospital → external scheduling service: Appointment slot request
MF_06	external scheduling service → hospital: Appointment slot response
MF_07	hospital → external correspondence service: Appointment confirmation letter
MF_08	hospital → patient: Telephone contact attempt
MF_09	patient → hospital: Patient contact outcome
MF_10	hospital → external clinical services: Treatment capacity request
MF_11	external clinical services → hospital: Treatment capacity response
MF_12	hospital → funding organisation: Funding approval request
MF_13	funding organisation → hospital: Funding approval response
MF_14	hospital → payment service provider: Secure payment request
MF_15	payment service provider → hospital: Payment status

Backup

B How the model is executed (for code questions)

AbstractWorker.java	the single place that decides how a failure reaches the engine
JobContext.java	reads and writes process variables; refuses blank or wrongly typed input
WorkerConfig.java	configuration precedence: environment → properties file → classpath → defaults
WorkersApplication.java	starts the fleet; refuses to start if a declared job type has no subscriber
tasks/*Worker.java (11 files)	one worker per Zeebe job type declared in the BPMN service and send tasks
support/ExternalServices.java	the simulated providers: scheduling, correspondence, clinical, payment
support/HospitalStore.java + OperationsLog.java	in-memory episode records and the audit entries the workers file
src/test (4 files, 41 tests)	worker contract tests — evidence that every job type is subscribed and its variables behave

Full element-by-element mapping for all four models (lane attribution, task identifiers, gateway conditions, form references and job types): 08_Repository_Documents/model-inventory.md

Deployment: 11 of 11 job types have a subscriber; 10 ran live and one is bound to a monthly timer-start process, verified structurally.